

EPT-TEST-15(IIIT Hyderabad Full Length Test - 2)

Question Paper Contain

50 Aptitude | 8 Algorithms | 9 C-Programming & DS |

11 Maths | 7 CN | 6 DBMS | 5 OS | Digital 4 |

Total 100 Marks | 180 Minutes Duration

+1 Mark to each correct answer

-0.25 Mark to each wrong answer

Q1) Consider the following inequalities.

A. $2x-1>7$

B. $2x-9<1$

Which one of the following expressions below satisfies the above two inequalities?

A. $x\leq-4$

B. $-4<x\leq4$

C. $4<x<5$

D. $x\geq5$

2.) Six students P, Q, R, S, T and U, with distinct heights, compare their heights and make the following observations.

1.) Observation I: S is taller than R.

2.) Observation II: Q is the shortest of all.

3.) Observation III: U is taller than only one student.

4.) Observation IV: T is taller than S but is not the tallest

The number of students that are taller than R is the same as the number of students shorter than _____.

A. T

B. R

C. S

D. P

Q3.) The ratio of boys to girls in a class is 7 to 3.

Among the options below, an acceptable value for the total number of students in the class is:

- A. 21**
- B. 37**
- C. 50**
- D. 73**

Q4.) Consider the following sentences:

- Everybody in the class is prepared for the exam.**
- Babu invited Danish to his home because he enjoys playing chess.**

Which of the following is the CORRECT observation about the above two sentences?

- A. (i) is grammatically correct and (ii) is unambiguous**
- B. (i) is grammatically incorrect and (ii) is unambiguous**
- C. (i) is grammatically correct and (ii) is ambiguous**
- D. (i) is grammatically incorrect and (ii) is ambiguous**

Q5.)_____ is to *surgery* as *writer* is to _____

Which one of the following options maintains a similar logical relation in the above sentence?

- A. Plan, outline**
- B. Hospital, library**
- C. Doctor, book**
- D. Medicine, grammar**

Q6.) We have 2 rectangular sheets of paper, M and N, of dimensions $6\text{ cm} \times 1\text{ cm}$ each. Sheet M is rolled to form an open cylinder by bringing the short edges of the sheet together. Sheet N is cut into equal square patches and assembled to form the largest possible closed cube. Assuming the ends of the cylinder are closed, the ratio of the volume of the cylinder to that of the cube is _____.

- A. $\pi/2$**
- B. $3/\pi$**
- C. $9/\pi$**
- D. 3π**

Q7.) In rolling two dice, find the probability that the sum is 7?

- A $1/4$**
- B $1/9$**
- C $1/2$**
- D $1/6$**

Q8.) When the trade was brisk. He worked hard and made his fortune; he believes in making hay while the sun shines.

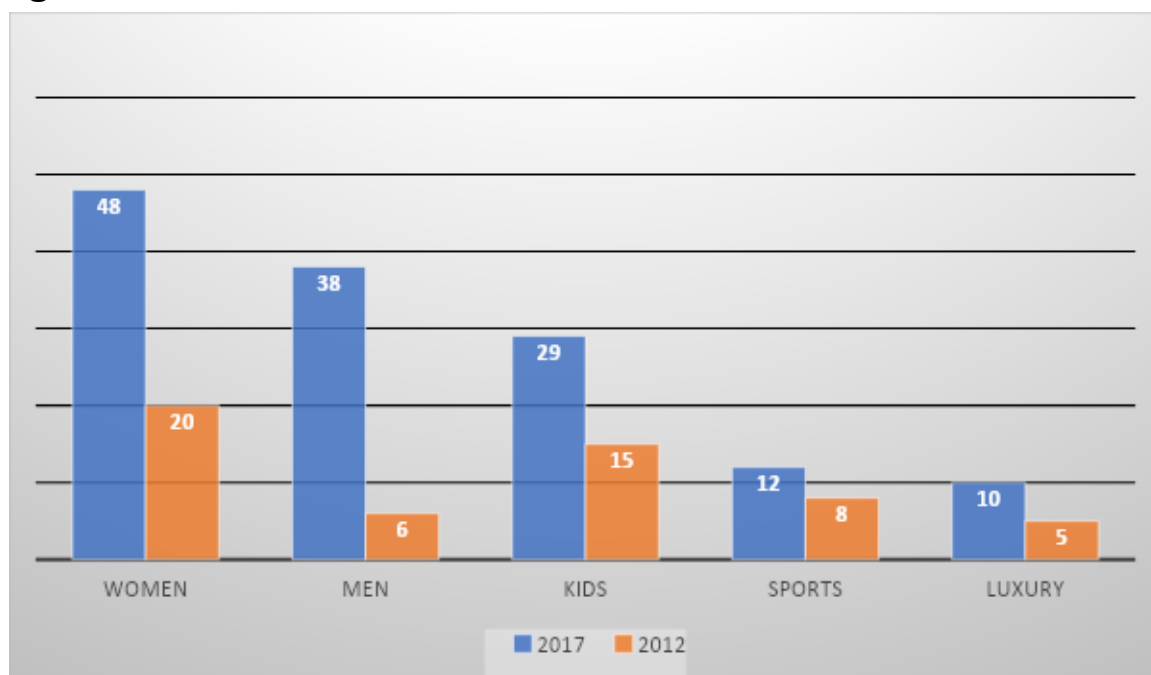
Choose the option which is closest in meaning to the underlined phrase in the above sentence.

- A Taking advantage of a favorable opportunity**
- B Earning money at the cost of others**
- C Taking advantage of the inflationary trends**
- D Earning money through dishonest means**

**Q9.) There are five bags each containing identical sets of ten distinct chocolates. One chocolate is picked from each bag.
The probability that at least two chocolates are identical is**

-
- A. 0.3024**
 - B. 0.4235**
 - C. 0.6976**
 - D. 0.8125**

Q 10.) A multi-national conglomerate sells soap products for five different market segments, namely (i) Women, (ii) Men, (iii) Kids, (iv) Sports, and (v) Luxury. The sales of these five segments (in lakh number of packs) during 2012 and 2017 are shown in the following figure.



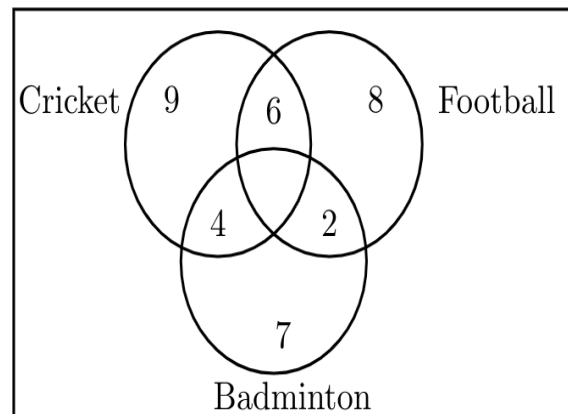
By what percentage were the sales of the Women segment in 2017 more than the sales of the Men segment in 2017?

- A. 29.67**
- B. 26.32**
- C. 25.33**
- D. 29.89**

Q 11.) In a class there are 40 students who play at least one game out of Football, Cricket and Badminton.

What percentage of students play only one game?

- A. 50%**
- B. 60%**
- C. 65%**
- D. 70%**

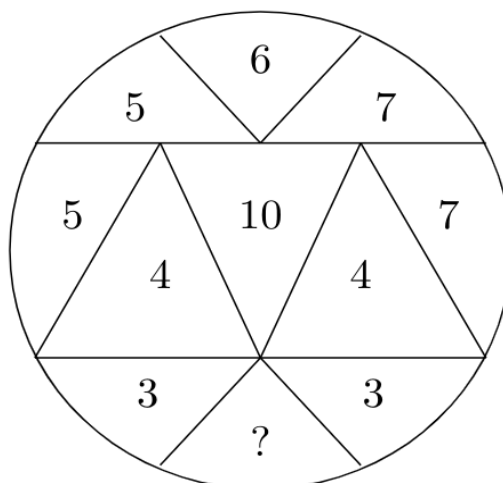


Q12.) Five people are standing in a row. VIRAT is standing next to PANDYA but not adjacent to SURYA. ISHAN is standing next to ROHIT who is standing on the extreme left and SURYA is not standing next to ISHAN. Who are Standing adjacent to VIRAT?

- A. ISHAN and PANDYA**
- B. PANDYA and SURYA**
- C. PANDYA and ROHIT**
- D. ISHAN and SURYA**

Q 13.) Find the missing number.

- A. 14**
- B. 10**
- C. 9**
- D. 3**



Q 14.) A businessman purchases an item at a certain price and marks its price up by 30%. He sells the item at a certain discount on

markup price and makes a net profit of 4% on the whole transaction. Find the discount given by businessman on markup price.

- A. 10
- B. 15
- C. 26
- D. 20

Q15.) Considering 5 as the 1st element in the sequence 5, 11, 23, 47. What is the 6th element in the sequence?

- A. 191
- B. 172
- C. 342
- D. 106

Q16.) Ram can do a piece of work in 5 days, and Sham can do the same in 10 days. With the help of Karan, they finished the work in 2 days. How many days would it take Karan to do the work?

- A. 5 days
- B. 10 days
- C. 15 days
- D. 20 days

Q17.) Find the mode of the following data:

- A. 20.22
- B. 19.47
- C. 21.12
- D. 20.14

AGE	0-6	6-12	12-18	18-24	24-30	30-36	36-42
FREQUENCY	6	11	25	35	18	12	6

Q18.) Study the following information carefully and answer the question:

Group of cricketers' gossip with each other. All are sitting surrounding a round table. The name of the cricketers are Rohit, Ishan, Pandya, sky and Bumra. It is not necessary that they are sitting in the order of the name as mentioned here. Ishan is second to the right of Rohit. Rohit doesn't sit with Pandya. Bumra is second to the right of Ishan. Ishan sits near sky.

Who sits to the left of Rohit?

- A. Bumra
- B. Ishan
- C. Pandya
- D. Sky

19.) Two statements followed by four conclusions numbered from (I) to (IV) are given. You have to take the two statements to be true even if these seem to be at variance from the commonly known facts. Read all the conclusions and decide which of the given conclusions logically follow from the two given statements disregarding commonly known facts.

- a. All Boys are Girls.
- b. No Girl is a Man.

Conclusions:

- I. No Boy is a Man
- II. Some Boys are Man
- III. All Girls are Boys
- IV. Some Man are Boys

- A. Only (III) follows
- B. Only (I) follows
- C. All follows
- D. None follows

20.) Two statements followed by four conclusions numbered from (I) to (IV) are given. You have to take the two statements to be true even if these seem to be at variance from the commonly known

facts. Read all the conclusions and decide which of the given conclusions logically follow from the two given statements disregarding commonly known facts.

- **All Sentences are Words**
- **All Words are Alphabets**

Conclusions:

- I. All Words are sentences**
 - II. All Sentences are alphabets**
 - III. All alphabets are words**
 - IV. Some alphabets are words**
- A. Only (I) and (III) follows**
 - B. Only (II) , (III) and (IV) follows**
 - C. Only (II) and (IV) follows**
 - D. All follows**

21.) Find the number which does not fit into the series 8 12 20 32 50 68.

- A. 20**
- B. 32**
- C. 68**
- D. 50**

Q.22) 5 16 49 104 181 _____

- A. 271**
- B. 298**
- C. 280**
- D. 281**

Q23.)14,28,20,40,32,64

- A. 52**

- B. 56**
- C. 96**
- D. 128**

Q24.) The students of a school participates in various sports activities, the distribution of the same is given below:

- Football–17%**
- Handball–26%**
- Badminton–16%**
- Table Tennis–22%**
- Basketball–19%**

Total number of students in the school are 800.

What is the respective ratio between the total number of students taking part in Badminton and Table Tennis together and those participating in Basketball and Football together?

- A. 11:13**
- B. 18:19**
- C. 19:18**
- D. 29:28**

Q 25.) The following table shows total number of students in the Department of the Institute along with percentage of Females and Male students. Answer the question based on the data given below:

What is the total number of students in all the Departments?

- A. 2210
- B. 3210
- C. 3400
- D. 3310

DEPART MENT	TOTAL NO OF STUDENTS	% OF FEMALE	%OF MA
COMPUTER	840	45	55
CIVIL	220	35	66
MECH	900	23	77
ELECTRICAL	360	65	35
ELECTRONICS	450	44	56
MINING	540	40	60

Q26.) Rajiv Gandhi Khel Ratna Award was conferred _____ Mary Kom, a six-time world champion in boxing, recently in a ceremony _____ the Rashtrapati Bhawan (the President's official residence) in New Delhi.

- A. with, at
- B. on, in
- C. on, at
- D. to, at

Q 27.) For a matrix $M=[m_{ij}]; i,j=1,2,3,4$, the diagonal elements are all zero and $m_{ij}=-m_{ij}$. The minimum number of elements required to fully specify the matrix is _____

- A. 0
- B. 6
- C. 12
- D. 16

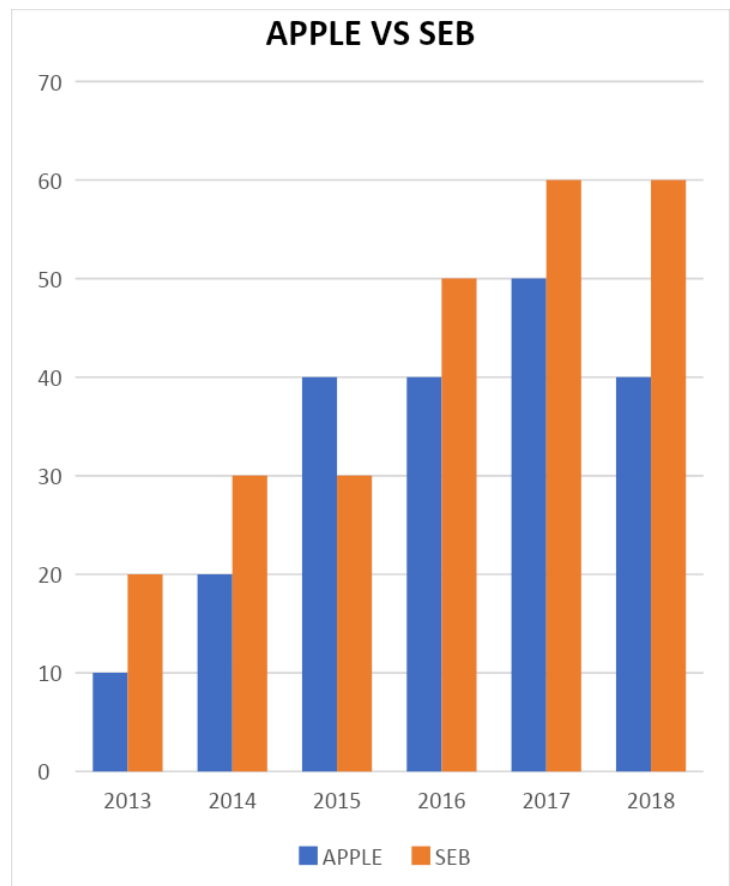
Q28.) Repo rate is the rate at which Reserve Bank of India (RBI) lends commercial banks, and reverse repo rate is the rate at which (RBI) borrows money from commercial banks.

Which of the following statements can be inferred from the above passage?

- A. Decrease in repo rate will increase cost of borrowing and decrease lending by commercial banks.**
- B. Increase in repo rate will decrease cost of borrowing and increase lending by commercial banks.**
- C. Increase in repo rate will decrease cost of borrowing and decrease lending by commercial banks.**
- D. Decrease in repo rate will decrease cost of borrowing and increase lending by commercial banks.**

Q29.) The profit shares of two companies APPLE and SEB are shown in the figure. If the two companies have invested a fixed and equal amount every year, then the ratio of the total revenue of company **APPLE to the revenue of company **SEB**, during 2013–2018 is**

- _____.
- A. 15:17**
 - B. 16:17**
 - C. 17:15**
 - D. 17:16**



Q30.) While I agree _____ her proposal this time, I do not often agree _____ her.

- A. to, with**
- B. with, to**
- C. with, with**
- D. to, to**

Q31.) Select the word that fits the analogy:

White: Whitening : : Light : _____

- A. Lightning**
- B. Lightening**
- C. Lighting**
- D. Enlightening**

Q32) In one of the greatest innings ever seen in 142 years of Test history, Ben Stokes upped the tempo in a five-and-a-half hour long stay of 219 balls including 11 fours and 8 sixes that saw him finish on a 135 not out as England squared the five-match series.

Based on their connotations in the given passage, which one of the following meanings DOES NOT match?

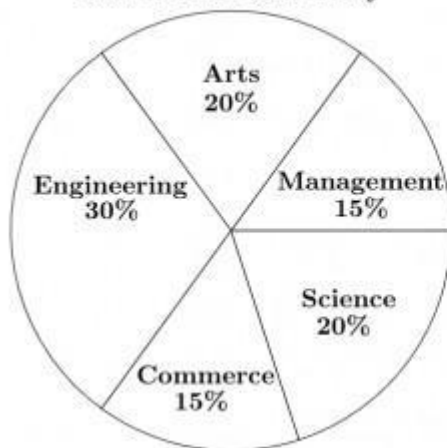
- A. upped = increased**
- B. squared = lost**
- C. tempo = enthusiasm**
- D. saw = resulted in**

Q 33.) IT was estimated that 52 men can complete a strip in a newly constructed highway connecting cities MUMBAI and PUNE in 10 days, Due to an emergency, 12 men were sent to another project. How many numbers of days, more than the original estimate, will be required to complete the strip?

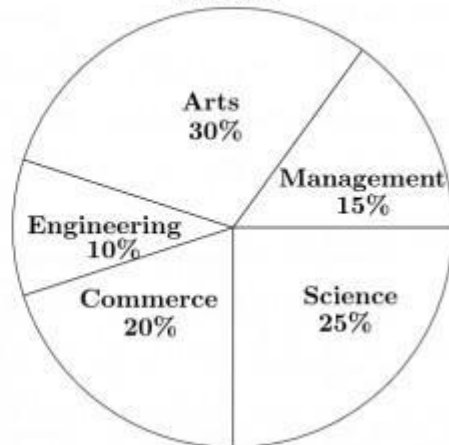
- A. 3 days**
- B. 5 days**
- C. 10 days**
- D. 13 days**

Q34.) The two pie-charts given below show the data of total students and only girls registered in different streams in a university. If the total number of students registered in the university is 5000, and the total number of the registered girls is 1500; then, the ratio of boys enrolled in Arts to the girls enrolled in Management is _____

Percentage of students enrolled in different streams in a University



Percentage of girls enrolled in different streams



- A. 2:1**
- B. 9:22**
- C. 11:9**
- D. 22:9**

Q35.) Consider the following three statements.

- I. Meera is older than Chandani**
- II. Pooja is older than Meera.**
- III. Chandani is younger than Pooja.**

If the first two statements are true, the third statement is?

- A. Cannot be determined**
- B. False**
- C. True**
- D. May be True**

Q 36.) In a family of seven people Lawyer is married to a Teacher and has three sons, one Engineer, one Doctor and one Actor. The Actor's wife is a Dancer and is an Aunt of Emily. Emily, the daughter of Engineer learns martial arts with her brother Rahul. How is the Doctor related to Rahul?

- A. Uncle**
- B. Son**
- C. Nephew**
- D. Brother**

37.) Find the smallest number x such that $x \times 162$ (x multiplied by 162) is a perfect cube

- A. 24**
- B. 27**
- C. 36**
- D. 38**

Q 38.) What is the maximum number of distinct handshakes that can happen in the room with 5 people in it?

- A. 15**
- B. 10**
- C. 6**
- D. 5**

Q39.) The present ages of three persons in proportions 4:7:9. Eight years ago, the sum of their ages was 56. Find their present ages (in years).

- A. 8,20,28**
- B. 16,28,36**
- C. 20,35,45**
- D. None of the above options**

Q40.)

This book, including all its chapters, _____ interesting. The students as well as the instructor _____ in agreement about it.

- A. is, was**
- B. are, are**
- C. is, are**
- D. were, was**

Q 41.) The American psychologist Howard Gardner expounds that human intelligence can be subcategorized into multiple kinds, in such a way that individuals differ with respect to their relative competence in each kind. Based on this theory, modern educationists insist on prescribing multi-dimensional curriculum and evaluation parameters that enable development and assessment of multiple intelligences.

Which of the following statements can be inferred from the given text ?

- A. Howard Gardener insists that the teaching curriculum and evaluation needs to be multi-dimensional.**
- B. Howard Gardener wants to develop and assess the theory of multiple intelligences.**
- C. Modern educationists want to develop and assess the theory of multiple intelligences.**
- D. Modern educationists insist that the teaching curriculum and evaluation needs to be multi-dimensional.**

Q42.) Insert seven numbers between 2 and 34, such that the resulting sequence including 2 and 34 is an arithmetic progression. The sum of these inserted seven numbers is _____.

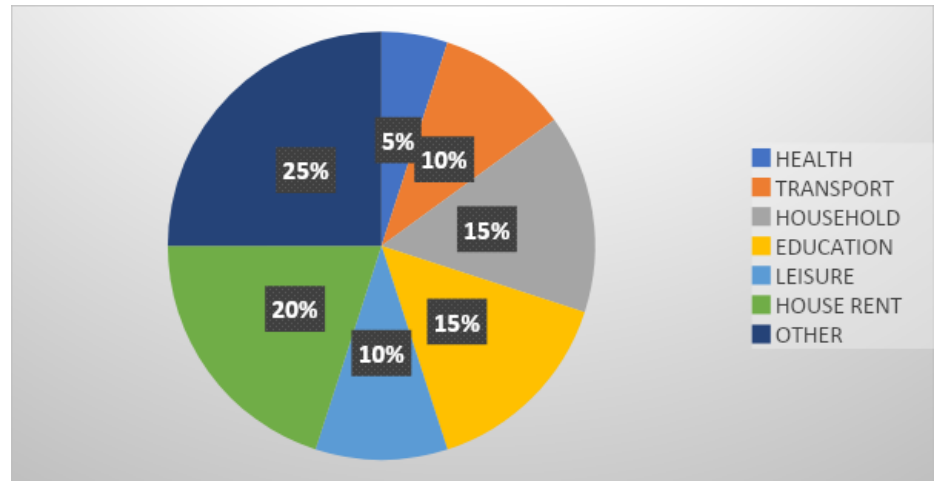
- A. 120**
- B. 124**
- C. 126**
- D. 130**

Q 43.) The unit's place in 26591749^{110016} is _____.

- A. 1**
- B. 3**
- C. 6**
- D. 9**

Q 44.)The total expenditure of a family, on different activities in a month, is shown in the pie-chart. The extra money spent on education as compared to transport (in percent) is _____.

- A. 5
- B. 33.3
- C. 50
- D. 100



Q 45.) He is known for his unscrupulous ways. He always sheds _____ tears to deceive people.

- A. fox's
- B. crocodile's
- C. crocodile
- D. fox

46.) In a class interval (11-15), the size of the class interval is?

- A 5
- B 15
- C 11
- D 4

47.) In rolling two dice, find the probability that the sum is 5?

- A $\frac{1}{4}$
- B $\frac{1}{9}$
- C $\frac{1}{2}$
- D $\frac{1}{6}$

48.) From the choices given below, select the pair of words which exhibits the same relationship between each other as the given capitalized pair of words TRIANGLE : PRISM ::

A circle : cylinder

B square : rhomboid

C sphere : earth

D polygon : diamond

Q49.) The recent measures to improve the output would _____ the level of production to our satisfaction

A. increase

B. decrease

C. speed

D. equalize

50.) 'Poverty is ... more restrictive and limiting than anything else. If poverty and low standards continue then democracy, for all its fine institutions and ideals, ceases to be a liberating force. It must, therefore, aim continuously at the eradication of poverty and its companion unemployment. In other words, political democracy is not enough. It must develop into economic democracy also' Which of the following is not implied by the above passage?

A Democracies should aim to eliminate poverty

B Democracy has ceased to be a liberating force

C Poverty and unemployment go hand in hand

D Political democracy should develop into economic democracy

Q51.) #include<stdio.h>

int main()

{

int x, y;

for(x=5;x>=1;x--)

{

for(y=1;y<=x;y++)

{

printf("%d\n",y);

}

}

return 0;

}

How many values printed by the below program ?

A 15

B 11

C 10

D 13

Q52.)

$$\lim_{x \rightarrow \infty} \left(\frac{x}{1+x} \right)^x$$

A 0

B 1

C e

D 1/e

Q53.) #include<stdio.h>
int main()
{
int y[10], i, *p=&i;
for(i=0;i<10;i++)
y[i]=i+1;
p= y;
p = p + 2;
*p=i;
printf("%d \n",y[2] * 4[y]+2);
}

- A. 51
- B. 52
- C. 53
- D. 54

Q54.) Consider the set $A = \{3, \mu, -1, \{2, 5\}\}$
The cardinality of the power set of A is _____.

- A. 4
- B. 8
- C. 16
- D. 32

Q55.) In the Stop and wait protocol, the link utilization is 40%, if the bandwidth 100 KB/sec. If we are using a frame size of 1 KB, what is the propagation time in milli sec with the basic stop-and-wait protocol?

- A. 7 to 7.2
- B. 7.5 to 7.7
- C. 8 to 8.2
- D. 9 to 9.2

Q56.) For the given relation $R(A_1, A_2, \dots, A_n)$. What is the number of super keys if candidate keys are $\{(A_1, A_2), (A_2, A_4)\}$?

A $2^{n-2} + 2^{n-2} + 2^{n-3}$

B $2^{n-2} + 2^{n-2}$

C $2^{n-2} + 2^{n-2} - 2^{n-3}$

D. 2^{n-2}

Q57.) What is the cardinality of the relation in between Course and Student in the given diagram?



A A Course could be enrolled by at most one Student and a Student can enroll in at most one Course.

B A Course could be enrolled by M number of Students and a Student can enroll in at most one Course.

C A Course could be enrolled by at most one Student and a Student can enroll in M number of Courses.

D A Course could be enrolled by M number of Students and a Student can enroll in M number of Courses.

Q58.) Consider a singly linked list of length n ($n \gg 1000$), having a first and last pointer which of the following operations do not depend on the length of the list.

A Printing the 100th element from the head.

B Deleting an element at the end of the list.

C Printing the middle two or single middle element of the list.

D Interchanging the first and the second last element of the list.

Q59.) The number of ways in which a student can choose 5 courses in a semester out of 9 courses if 2 of them are compulsory.

A 32

B 33

C 34

D 35

Q60.) #include <stdio.h>

int a=0;

int test(int i)

{

int x = 0,j;

for (j =1;j<3;j++)

{

x = x + i%10;

i = i/10;

a= a+x;

}

return a;

}

int main()

{

printf("%d\n",test(24));

return 0;

}

A. 10

B. 11

C. 12

D. 13

Q61.) Dialing a telephone number a person the last three digits for his friends number, he randomly dials the last three digits of the number the probability that the number is correct is

- A $1/10$
- B $1/1000$
- C $1/125$
- D $1/720$

Q62.) A company wants to create the 6 subnets with network address 192.123.220.0/23. Each subnet must have a maximum number of hosts possible. What is the address of the fourth host of the third subnet.

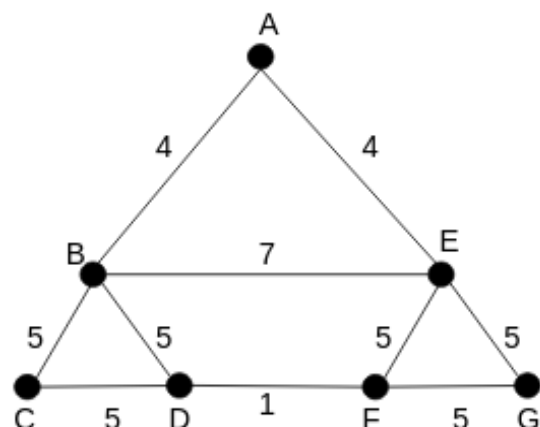
- A 192.123.220.68
- B 192.123.220.132
- C 192.123.221.132
- D None of these

Q63.) $|1\ 1\ 1\ 1\ x + 1\ 1\ 1\ 1\ y + 1| =$

- A. $x+y$
- B. $x-y$
- C. xy
- D. x^2y^2

Q64.) Consider the the graph given below the cost of the minimal spanning tree of the below graph is

- A. 24
- B. 12
- C. 16
- D. 18



Q65.) S1 :The function $f: \mathbb{N} \rightarrow \mathbb{R}$, defined by $f(n) = -\sqrt{n}$, is an injection (1-1 function) but not a surjection (onto function)

S2 : Every one-to-one function has an inverse function.

Note : N is the set of natural numbers and R is the set of real numbers.

A S1 is True, S2 is False

B S1 is False, S2 is True

C Both S1 and S2 are True

D Both S1 and S2 are False

Q66.) Match the following according to their correct port numbers

Column B	
i	52
ii	53
iii	21
iv	80
v	110
vi	25
COLUMN A	
1	SMTP
2	HTTP
3	DNS
4	POP client
5	FTP

A 1-(vi), 2 - (iv), 3- (ii), 4- (v), 5- (iii)

B 1-(iii), 2 - (iv), 3- (i), 4- (v), 5- (vi)

C 1-(iii), 2 - (v), 3- (ii), 4- (iv), 5- (vi)

D 1-(vi), 2 - (iv), 3- (i), 4- (v), 5- (iii)

QQ67.) The grey code of $(B6F)_{16}$ could be represented as?

A. $(F58)_{16}$

B. $(E55)_{16}$

C. $(E5F)_{16}$

D. $(ED8)_{16}$

Q68.) Which of the following is not an inplace sorting algorithm.

- A Merge Sort
- B Quick Sort
- C Insertion Sort
- D Bubble sort

Q69.) Evaluate the following expression

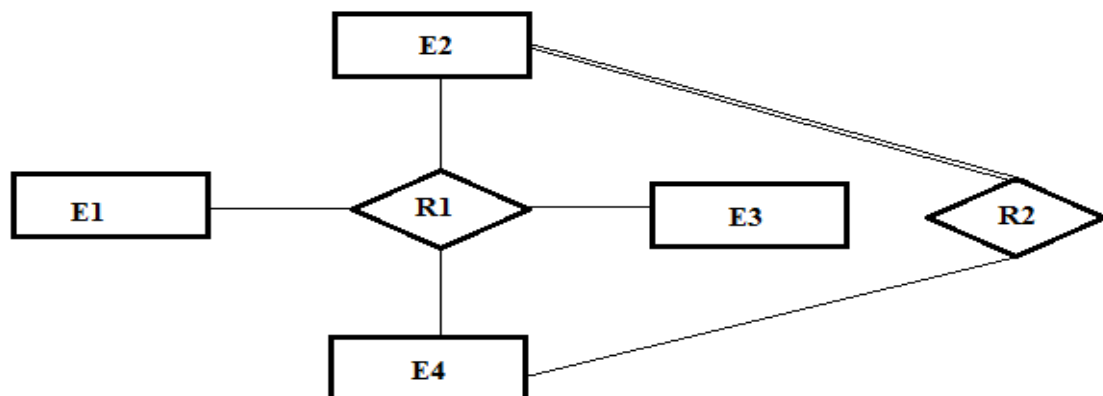
$27 \cdot 27 \cdot 2 + 3 \cdot 4 + 12 \cdot 7 + 3 \cdot 12 \cdot 2^* -$

- A. -255
- B. 255
- C. 127
- D. -127

Q70.) Consider the below given ER diagram?

The minimum number of relations required for the given ER diagram are _____

NOTE: We can consider the composite primary keys.



- A. 2
- B. 5
- C. 4
- D. 3

Q71.) Assume a file system with the following layout and sizes:
blocks: 512 bytes,

**inodes: 64 bytes,
directory entries: 16 bytes,
inode numbers: 2bytes, and
block numbers: 4 bytes**

The file system has 12 direct pointers, 1 indirect pointer in the i-node.

Assume only a single directory (the root).

What is the maximum size a file can be?

A 68740 bytes

B 71680 bytes

C 74770 bytes

D None of the above

Q72.)Which of the following data structures is used for breadth first search.

A Stack

B Queue

C Heap

D Circularly linked list

Q73.)Consider classic dinning philosopher problem with arbitrary number of D diners. Furthermore, each diner $d=1 \dots D$ may require a different number of S_d spoons to eat. For example, it is possible

that one of the diners is an octopus, who for some reason refuses to begin eating before acquiring $S_{octopus} = 8$ spoons. What is the smallest number of spoons needed to ensure that deadlock can not occur?

A. $\sum_d (S_d - D + 1)$

B. $\sum_d (S_d + D)$

C. $\sum_d (S_d - 1)$

D. $\sum_d (S_d - D)$

Q74.) What is the content of array A after executing the following code snippet?

```
int A[3] = {1, 2, 3};
int *p;
int **q;
p = A;
p++;
q = &p;
p++;
(*p) = (**q)*2;
```

- A. 123
- B. 124
- C. 126
- D. 143

Q75.) The K-cube is the graph whose vertices are the ordered K-tuples of 0's and 1's, two vertices being adjacent if and only if they differ in exactly one coordinate(i.e. bit). For example, (000) and (001) are adjacent But (000) and (011) are not.

Consider the following statements about K-cube graph :

- **S1: It has 2^K vertices**
- **S2: It has $K \times 2^{(K-1)}$ edges**
- **S3: It is a bipartite graph**

Which one of the following statements is true?

- A. Only S1**
- B. Only S1, S2**
- C. Only S1, S3**
- D. All of S1, S2, and S3**

Q76.) Given a set of values $R=\{1,2,3,4,5,6,7\}$, What is the number of relations on this set which is both partial-order and equivalence relation?

- A. 0**
- B. 1**
- C. 2**
- D. 3**

77.) Consider a file system with 2048 byte blocks and 32-bit disk. How large of a disk (in Tera bytes) can this file system support?

- A. 8**
- B. 9**
- C. 10**
- D. 12**

Q78.) consider that nodes in slotted-Aloha transmit in each slot with probability p. Suppose that a given channel has

exactly 8 nodes. What is the probability that there is a collision in a given slot?

- A. $1-(1-p)^8$
- B. $1-8p(1-p)^7-(1-p)^8$
- C. $1-8p(1-p)^7$
- D. $(1-p)^8$

79.) Suppose we have an undirected, connected, weighted graph with n vertices such that:

- Every weight is an integer greater than zero.
- No two edges have the same weight.

What is the lowest cost a minimum spanning tree (MST) could have for a graph with 20 nodes that meets the description above?

- A. 150
- B. 170
- C. 180
- D. 190

80.) Let E and F be two independent events, with $P(E) = P(F) = 0.25$. Let G be another event on the same sample space, which is disjoint from both E and F , and with $P(G) = 0$. Then, $P(G|F)$ is

- A. 1
- B. 0.5
- C. 0.25
- D. 0

81.) A decoder may be described as

- A. Multiple-input single-output logic circuit which converts coded inputs into a coded output where the input and output codes are different.**
- B. Single-input single-output logic circuit which converts the coded input into a coded output where the input and output codes are different.**
- C. Multiple-input multiple-output logic circuit which converts coded inputs into coded outputs where the input and output codes are different.**
- D. Single-input multiple-output logic circuit which converts the coded input into coded outputs where the input and output codes are different.**

Which of the option is correct

- A. A**
- B. B**
- C. C**
- D. D**

82.) Suppose $g(x)$ is a polynomial function such that $g(-1)=4$ and $g(2)=7$. Then there is a number c between -1 and 2 such that

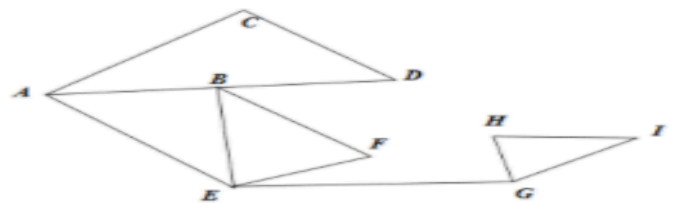
- A. $g(c)=1$**
- B. $g(c)=0$**
- C. $g'(c)=1$**
- D. $g'(c)=0$**

Q83.) Which of the following can't be the contents of the stack from bottom to top at a time instant in evaluating the postfix expression : 8 6 3 / + 3 2 * -
(where * is the multiplication operator)?

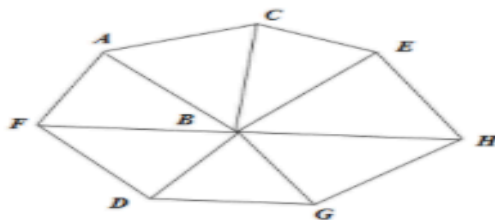
- A. 10,6
- B. 10,3
- C. 10,3,2
- D. 10,5

Q.84) Consider the two graphs
Which of the following graphs have a Hamiltonian circuit

Graph A :



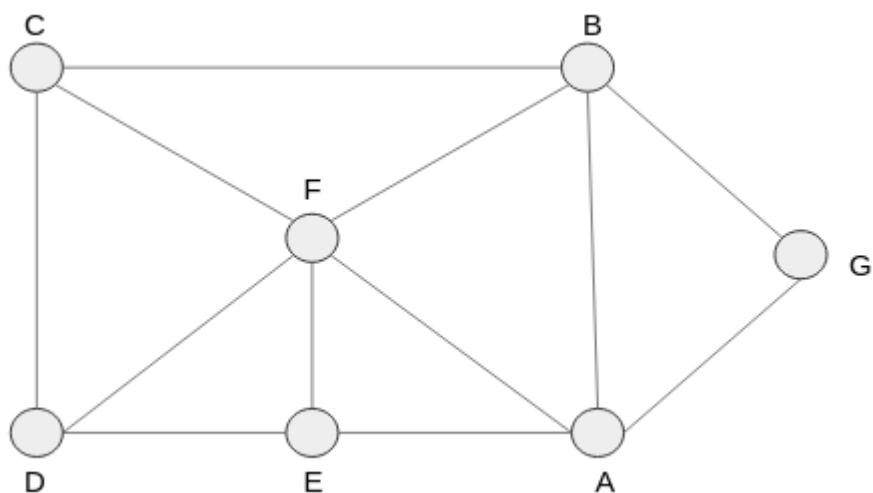
Graph B :



- A G1 only
- B G2 only
- C Both G1 and G2
- D None of the above

Q85.) Which of the following is not a valid DFS traversal of the graph.

- A.) C, B, F D, E, A, G
- B.) C, D, F, B, G, A, E



C.) A, G, B, D, E, F, C

D.) F, C, B, G, A, E, D

Q86.) two of the following are true regarding the distance-vector and link-state routing protocols?

A Link state sends its complete routing table out all active interfaces on periodic time intervals.

B Distance vector sends its complete routing table out all active interfaces on periodic time intervals.

C Distance vector sends updates containing the state of its own links to all routers in the internetwork.

D None of the above

Which of the following is correct

A. A

B. B

C. C

D. D

Q87.) Consider a knapsack with capacity 20 units of weight and the following items with their respective profits, the optimal profit which can be obtained is , given that we can also add a fraction of items to the knapsack_____.

A. 40.5

B. 35.5

C. 37.5

D. 42.5

Item	Profit	Weight
1	20	10
2	8	16
3	7	1
4	5	40
5	10	2

88.) A block of addresses 205.16.37.128/28 is granted to a small organization. Organisation needs the 4 subnets with maximum hosts possible. What is the range of addresses in the last subnet?

A 205.16.37.128/28 - 205.16.37.128.255/28

B 205.16.37.128.140/30 - 205.16.37.255/30

C 205.16.37.140/30 - 205.16.37.143/30

D None of the above

Q89.) Arrange the following function in increasing order of time complexity

$f_1(n) = n^2 \log n$

$f_2(n) = (0.989)^n$

$f_3(n) = (1.000001)^n$

$f_4(n) = \log(\log(e^{e^n}))$

A 1, 4, 2, 3

B 1, 4, 3, 2

C 2, 4, 1, 3

D 4, 1, 2, 3

90.) Which of the following is the correct SOP expression equivalent to the POS expression $F = (A + B + C)(A + B' + C')$

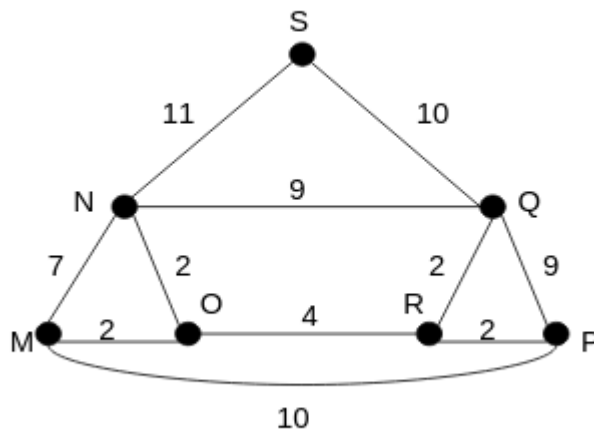
A $A'B'C + A'BC + AB'C + AB'C' + ABC' + ABC$

B $A'B'C' + A'BC + AB'C + AB'C' + ABC' + ABC$

C $A'B'C' + A'BC' + AB'C + AB'C' + ABC' + ABC$

D $A'B'C + A'BC' + AB'C' + AB'C + ABC' + ABC$

91.) Consider the following graph



Which of the following statements is true regarding the minimal spanning tree of the above graph.

- A The cost of the MST of the graph is 22
- B Edge MP is present as a part of every MST of the graph
- C Edge NM is present as a part of every MST of the graph
- D None of the above

92.) A machine has 48-bit virtual addresses and 32-bit physical addresses. For this byte addressable system, the Page Size is 8 KB. Which of the following is correct about the number of entries required for an inverted page table?

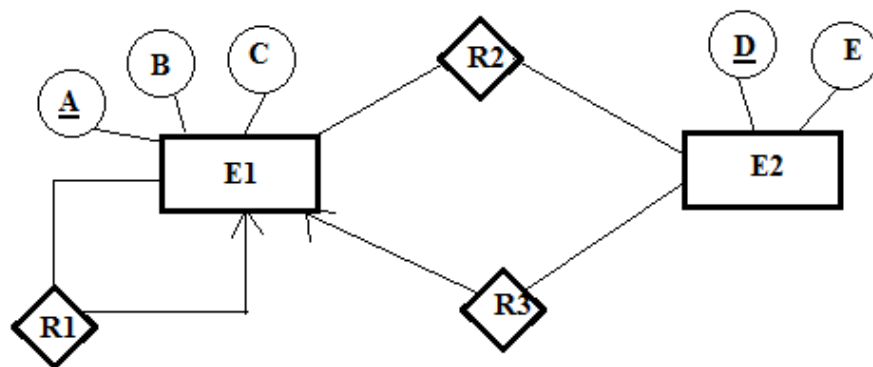
- A 2^{10}
- B 2^{15}
- C 2^{19}
- D 2^{22}

Q 93.) The maximum number of candidate keys for given 6 distinct attributes are _____

- A. 16
- B. 20
- C. 24
- D. 28

Q 94.)

Consider the below given ER diagram:



The number of foreign keys required for the above ER diagram are _____

- A. 4
- B. 5
- C. 3
- D. 2

Q 95.)Consider the relation

A->B

and

B->A

Which of the following relationship is correct for the two entities

A Many -to- Many

B Many -to- One

C One -to- Many

D One -to- One

96.)The maximum positive number represented using 8 bits in signed 2's complement is _____

A. 64

B. 128

C. 63

D. 127

97.)In a CSMA/CD network the length of the cable is 500 metre and cable capacity is 50 Mbps, the speed at which data transmits is 5×10^6 m/s. What is the minimum frame size in bytes for the CSMA/CD network?

A. 1100

B. 1250

C. 1400

D. 1600

```

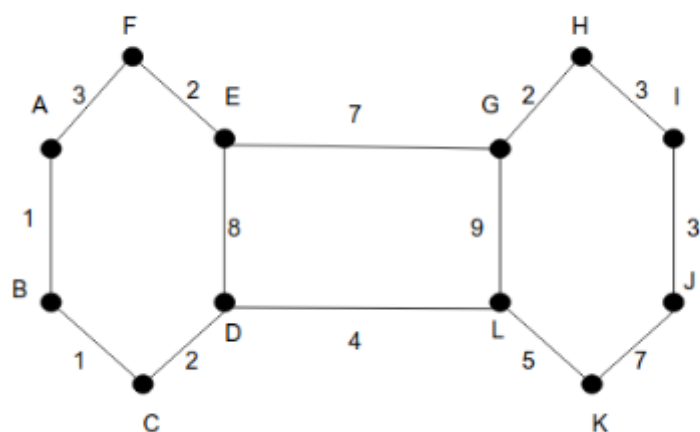
98.)#include<stdio.h>
#include<string.h>
int main ()
{

    char str[]= "COURSE";
    char *    ptr;
    ptr=str;
    ptr++;
    ++ *ptr;
    printf("%s",ptr); return 0 ;
}

```

A COURSE
 B OURSE
 C PURSE
 D URSE

99.) The cost of the
 MST for the following
 graph is ____



- A. 32
- B. 33
- C. 34
- D. 35

100.) Consider the following statements:

S1: Deadlock avoidance rarely used as a practical solution to the deadlock problem as it requires far too much overhead.

S2: If round-robin schedulers uses small size quantum, it improves the response time.

A S1 is true, S2 is false

B S1 is false, S2 is true

C Both S1 and S2 are true

D None of the above

1.) Solution C

Given that,

- $2x-1>7 \rightarrow (1)$
- $2x-9<1 \rightarrow (2)$

From equation (1),

$$2x-1>7$$

$$\Rightarrow 2x>8$$

$$\Rightarrow x>4$$

From equation (2),

$$2x-9<1$$

$$\Rightarrow 2x<10$$

$$\Rightarrow x<5$$

We can write, $4<x<5$

2.) Solution c

P (Tallest)

T

S

R

U

Q (shortest)

Number of students taller than R = 3

Number of students shorter than S = 3

3.) Solution C.

The number of boys is a multiple of 7 i.e., $7B$, for some integer B .

The number of girls is a multiple of 3 i.e., $3B$.

$\therefore$ The total number of students $= 7B + 3B = 10B$, i.e., a multiple of 10.

Only option that fits is C.

4.) Solution C

Everybody in the class is prepared for the exam.

The above statement is correct as when we say *everybody* or *everyone*, we are considering the class as a unit. So singular form of the verb “is” is correct.

- Babu invited Danish to his house because he enjoys playing chess.

The above statement is ambiguous because we are not sure who “he” refers to – it can be either Babu or Danish.

5.) Solution C

A doctor performs a surgery, and a writer writes a book.

6.) Solution c

Sheet M is rolled to form an open cylinder by bringing the shorter edge of the sheet together

So the longer side becomes the circumference of the base of the cylinder and shorter side becomes the height.

$$\therefore 2 \times \pi \times r = 6 \Rightarrow r = 3 / \pi$$

$$\text{Volume of cylinder} = \pi \times r^2 \times h = \pi \times (3 / \pi)^2 \times 1 = 9 / \pi$$

Sheet N is cut into equal square patches and assembled to form the largest possible closed cube

So the entire area of the sheet is converted into surface area of the cube, i.e.

$$6 \times a^2 = 6 \times 1 \Rightarrow a^2 = 1 \Rightarrow a = 1$$

$$\text{Volume of cube} = a^3 = (1)^3 = 1$$

$$\text{Volume of cylinder} / \text{Volume of cube} = (9 / \pi) / 1 = 9 / \pi$$

7.)Solution: (D)

On rolling two dice we have total outcomes as 36.

Sum of 5 can come in the following ways:

1+6, 2 +5, 3+4 or 4+3 or 5+2 or 6+1

Hence there are 6 ways to obtain a sum of 7.

So the required probability is $6/36 = 1/6$.

probability is $4/36 = 1/9$.

8.)Solution: (A)

Idiom 'make hay while the sun shines' means: to use favorable opportunities or conditions to our benefit/ to act when the time is right

9.) SOLUTION C

P(No two chocolates are

identical)= $10 \times 9 \times 8 \times 7 \times 6 / 10^5 = 30240 / 10^5 = 0.3024$

P(At least two chocolates are identical)= $1 - P(\text{No two chocolates are identical})$

= $1 - 0.3024 = 0.6976$

10.) SOLUTION B

= $(48 - 38) / 38 \times 100 = 26.32\%$

11.) SOLUTION B

The number of students, who play only one game is $= 9 + 8 + 7 = 24$

Percentage= $(24 / 40) \times 100\% = 60\%$

So, the correct answer is B

12.) SOLUTION A

VIRAT PANDYA or PANDYA VIRAT and it is given that VIRAT is not adjacent to SURYA.

ROHIT is standing on the Extreme left.

So ROHIT ISHAN _____ and SURYA is not standing next to ISHAN.

Now we can say that VIRAT is Left to PANDYA because if it right to PANDYA then SURYA will be adjacent to ISHAN or VIRAT but it is not possible.

So final positions would be

ROHIT– ISHAN–VIRAT – PANDYA – SURYA

According to question VIRAT is adjacent to ISHAN and PANDYA.

13)SOLUTION D

add numbers on both sides (L + R) then divide the sum by 2 to get the middle number.

$$(L+R)/2$$

$$\text{case 1- } (7+5)/2 = 6$$

$$\text{case 2- } (5+4+7+4)/2 = 10$$

$$\text{case 3- } (3+3)/2 = 3$$

14)Solution D

Let he purchased an item @ Rs.100

Marked price = Rs. 130

We know that

Marked price - Discount = Selling price

According To the Question

$$130 - 130 \times x \% = 104$$

$$130 - 13x/10 = 104$$

$$x = 20$$

therefore option D

15.) Solution A

Given that, 5,11,23,47, _____ , _____

We can do that, in this way,

A. $5 \rightarrow +6 = 11$

B. $11 \rightarrow +12 = 23$

C. $23 \rightarrow +24 = 47$

D. $47 \rightarrow +48 = 95$

E. $95 \rightarrow +96 = 191$

16.) SOLUTION A

Given that, R S K

Time: 5 10 X (days)

Total work : $\text{LCM}(5,10,x) = 50x$ unit.

Efficiency: $10x$ $5x$ 50

We know that, Total work = Efficiency $\times$ Time

$$\Rightarrow 50x = (15x + 50) \times 2$$

$$\Rightarrow 50x = 30x + 100$$

$$\Rightarrow 20x = 100$$

$$\Rightarrow x = 5 \text{ days.}$$

So, the correct answer is (A).

17.)Solution A

To find the mode of grouped data we have following formula:

$$\text{MODE} = l + \frac{(f_1 - f_0) / ((2 * (f_1)) - f_0 - f_2) * h}{1}$$

where l = lower limit of modal class

h = size of interval

f_1 = frequency of modal class

f_0 = frequency of class preceding the modal class

f_2 = frequency of class succeeding the modal class

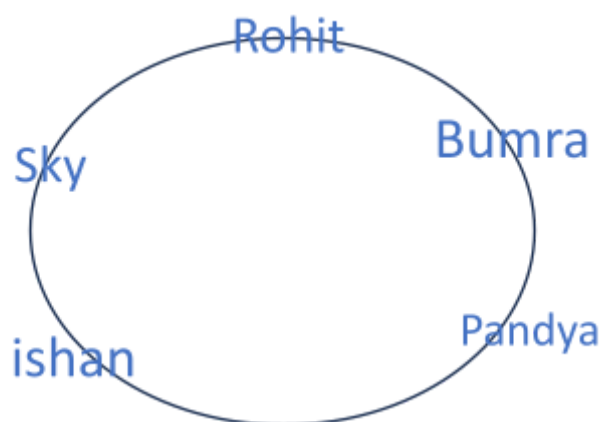
$$\therefore \Rightarrow \text{mode} = 18 + \frac{(35 - 25) / (2 * (35) - 25 - 18) * 6}{1}$$

$$\Rightarrow \text{mode} = 18 + (10 / 27) * 6$$

$$\Rightarrow \text{mode} = 20.22$$

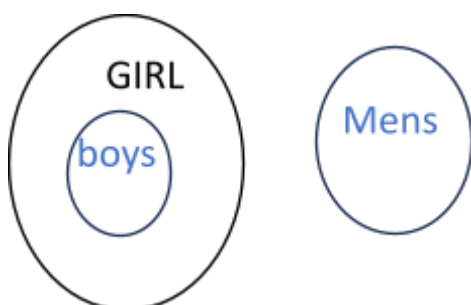
Option A is correct.

18.) Solution A

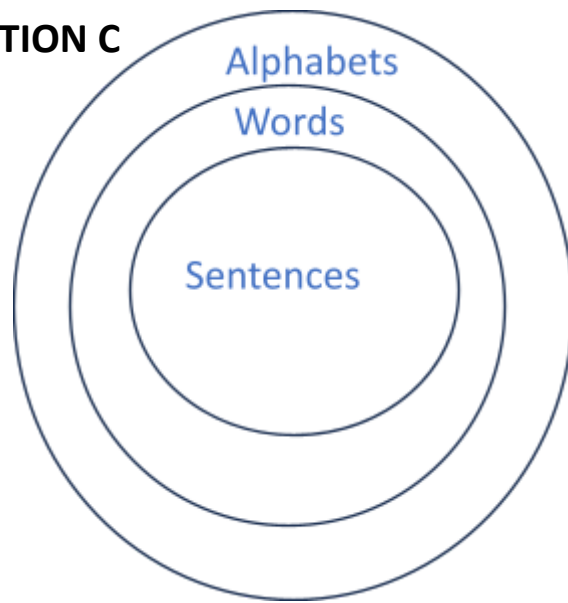


Left of Rohit option A) bumra

19.)Solution B



20.) SOLUTION C



21.) SOLUTION D

For given series pattern are as follows:

- A. $8+4*1=12$
- B. $12+4*2=20$
- C. $20+4*3=32$
- D. $32+4*4=48$
- E. $48+4*5=68$

Option D is correct.

22.) SOLUTION C

For given series pattern are as follows:

- $5+11=16$
- $16+33=49$
- $49+55=104$
- $104+77=181$
- $181+99=280$

Option C is correct.

23.)SOLUTION B

The pattern of the given series is as follows:

- $14 \times 2 = 28$
- $28 - 8 = 20$
- $20 \times 2 = 40$
- $40 - 8 = 32$
- $32 \times 2 = 64$
- $64 - 8 = 56$

Option (B) is correct.

24.) Solution c

Total number of students=800

Total number of students in Badminton= $16\%800 \Rightarrow 128$

Total number of students in Table tennis= $22\%800 \Rightarrow 176$

Total number of students in basketball= $19\%800 \Rightarrow 152$

Total number of students in football= $17\%800 \Rightarrow 136$

Required ratio= $128+176/152+136 \Rightarrow 304/288=19/18$

Option C is correct.

25 SOLUTION D

Option (D) is answer.

Total number of Students in all department =

$$840+220+900+360+450+540 = 3310$$

26 SOLUTION C

Conferred is followed by on/upon

27.) SOLUTION B

For a matrix $M=[m_{ij}]; i,j=1,2,3,4$, the diagonal elements are all zero and $m_{ij}=-m_{ji}$ (This is the property of skew-symmetric matrix).

Now, M

$$= \begin{bmatrix} 0 & -m_{21} & -m_{31} & -m_{41} \\ m_{21} & 0 & -m_{32} & -m_{42} \\ m_{31} & m_{32} & 0 & -m_{43} \\ m_{41} & m_{42} & m_{43} & 0 \end{bmatrix}$$

Out of 16 elements, 4 are 0s and 6 are the negative of the other 6. So, we need at least 6 elements to fully specify the matrix. So, the correct answer is (b).

28.) Solution D

“Repo rate is the rate at which Reserve Bank of India (RBI) lends commercial banks”

So a decrease in repo rate means commercial banks have to pay less.

29.)SOLUTION B

Let us assume both the companies APPLE and SEB invested Rs.100

Now, the total revenue of

$$\text{company APPLE} = 110 + 120 + 140 + 140 + 150 + 140 = 800$$

And, the total revenue of

$$\text{company SEB} = 120 + 130 + 130 + 150 + 160 + 160 = 850$$

$$\therefore \text{The required ratio} = 800:850 = 80:85 = 16:17.$$

30.)SOLUTION: A

"agree with" is used for referring to agreeing with a person

"agree to" is used for referring to agreeing with a non-person

Correct option: A.

31.) SOLUTION B

Whitening making a thing white.

Lightening makes a thing light.

32.) SOLUTION: B

Based on the given passage all other three usages are apt. The usage “squared” (earlier was behind) should mean that the series was tied after this match – either

33.) SOLUTION: A

52 men complete strip in 10 days

so 1 men will complete in 520 days

due to emergency 12 men sent so $52-12=40$ men

40 men will complete in $520/40=13$ days

estimate was 10 days

so the difference $13-10=3$ days. which is option A

34.) SOLUTION D

Given that, the total number of students registered in the university is 5000, and the total number of the registered girls is 1500.

- **Number of boys enrolled in**

Arts $= (5000 \times 20\%) - (1500 \times 30\%) = 1000 - 450 = 550$.

- **Number of girls enrolled in Management $= 1500 \times 15\% = 225$.**

Now, the required ratio $= 550:225 = 110:45 = 22:9$.

35.) SOLUTION: C

From the given statements:

Meera>Chandani and Pooja>Meera

From the above statements,

Pooja>Meera>Chandani

∴ the third statements should be true.

36.) solution A

Rahul is the son of the Engineer who is the brother of Doctor. So, the Doctor is Uncle of Rahul.

So, the correct answer is A

37.) SOLUTION C

36

$$162 = 2 * 9 * 9$$

$$36 = 2 * 2 * 9$$

38.) SOLUTION B

For maximum number of distinct handshakes, everyone should shake hands with everyone else exactly once. So, the number of handshakes will be same as the number of ways to choose two people from n people.

Hence, the maximum number of handshakes possible

$$\text{is: } nC_2 = \frac{n(n-1)}{2}$$

For n=5, the answer is $5C_2 = 10$.

39.) SOLUTION:B

Sum of age 8yrs ago was 56

Sum of age presently is $56 + 24 = 80$

Ratio is 4:7:9 at present.

$$4x + 7x + 9x = 20x \quad 20x = 80 \Rightarrow x = 4$$

So B is correct.

40.) SOLUTION C

is, are

- 'is' because "this book" is singular
- 'are' because for "as well as" the verb should agree with the noun preceding "as well as" which here is "the students".

41.) SOLUTION D

expounds: present and explain (a theory or idea) in detail.

A. Howard Gardener insists that the teaching curriculum and evaluation needs to be multi-dimensional.

Not true. "modern educationists insist on prescribing multi-dimensional"

B. Howard Gardener wants to develop and assess the theory of multiple intelligences.

Not true. "Howard Gardner expounds that human intelligence can be subcategorized into multiple kinds" meaning he explains the theory of multiple intelligence but we cannot infer if he wants to develop and assess that theory.

C. Modern educationists want to develop and assess the theory of multiple intelligences.

Not true. "Based on this theory, modern educationists insist on prescribing multi-dimensional curriculum" meaning modern educationists are using the theory – but we cannot infer if they want to develop it.

D. Modern educationists insist that the teaching curriculum and evaluation needs to be multi-dimensional.

True. This can be clearly inferred from "modern educationists insist on prescribing multi-dimensional curriculum"

42.) SOLUTION C

We can insert the seven numbers in such a way that, so we get the arithmetic progression including 2 and 34.

We get the arithmetic progression 2,6,10,14,18,22,26,30,34

The inserted seven numbers are 6,10,14,18,22,26,30

Here, $a=6, d=10-6=4, l=30, n=7$

$\therefore$ The sum of the arithmetic progression $S_7 = 7/2(6+30) = 7 \cdot 18 = 126$.

So, the correct answer is C

43.) SOLUTION A

We can write, the unit's place in $26591749^{110016} =$ the unit's place in 9^{110016}

Now, we know that the unit place of $9^1=9, 9^2=1, 9^3=9, \dots$

$\therefore$ The cyclicity of 9 is 2.

And, 110016 is divisible by 2, so the unit place of $9^2=1$.

44.) SOLUTION C

The expenditure of a family, on education =15%, on transport =10%.

The extra money spent on education as compared to transport $= ((15-10)/10) \times 100\% = 50\%$.

So, the correct answer is (C).

45.) SOLUTION C

Crocodile tears (or superficial sympathy) is a false, insincere display of emotion such as a hypocrite crying fake tears of grief.

So, the correct answer is C

46.) Solution A

Size of class interval is $(15-11+1)=5$

47.) SOLUTION B

On rolling two dice we have total outcomes as 36.

Sum of 5 can come in the following ways:

1+4, 2 +3, 3+2 or 4+1

Hence there are 4 ways to obtain a sum of 5.

So the required probability is $4/36 = 1/9$.

48.)Solution: (A)

Just as a prism is a solid figure which is triangular, a cylinder is a solid figure that is circular.

49.) SOLUTION: A

Measures are for improving the output. So, level of production is expected to “increase”.

50.) SOLUTION B

Solution: (B)

Solution: Option B is clearly not implied in the paragraph. The paragraph does talk about how it should democracy encompasses

51.) SOLUTION: A

X=5 y=1,2,3,4,5 X=4 y=1,2,3,4 X=3 y=1,2,3

$X=2 \ y=1,2 \quad X=1 \ y=1$
Total 25 value printed

52.) Solution d

This is 1^∞ indeterminate form

it can be solved using if $(f(x)^{g(x)})$ is in 1^∞ indeterminate form then the limit can be written as $(e^{g(x)(f(x) - 1)})$

it comes to $(e^{x(-x/(x+1))})$
 $=e^{-1}$

53.) SOLUTION B

Initially, p is holding the address of integer i.

Using for loop, the loop fails when $i=10$

The array $y[10] = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$

$p = y$ // p start pointing to array y.

$p = p + 2$ // p start pointing to the third location which has value 3.

$*p = i$ // 3 is replaced by 10

New array = $\{1, 2, 10, 4, 5, 6, 7, 8, 9, 10\}$

$y[2] = 10$

$4[y] = y[4] = 5$

$y[2] * 4[y] + 2 = (10 * 5) + 2 = 52$

54.) Solution C

$|P(A)| = 2^4 = 16$

55.) Solution: B

$T_t = 1 \text{ KB} / 100 \text{ KB/sec} = .01 \text{ sec}$

$0.4 = 1 / 1 + 2a$

$1 + 2a = 1 / 0.4$

$1 + 2 T_p / T_t = 1 / 0.4$

$T_p = 0.0075 \text{ sec} = 7.5 \text{ m sec}$

56.) SOLUTION C

For (A1, A2) => the number of superkeys will be 2^{n-2}

For (A1, A2) => the number of superkeys will be 2^{n-2}

For (A1, A2, A4) => number of subsets will be 2^{n-3}

Therefore, the number of keys will be: (A1,A2)+(A2,A4)-(A1,A2,A4)

=> $2^{n-2} + 2^{n-2} - 2^{n-3}$

57.) Solution: B

The relationship “Enroll in” is one-to-many + both side partial participation between the entity Course and Student. Therefore, any number of students could be enrolled in a course but a student can enroll in only one course. Therefore, option B is correct.

58.) Solution A

Printing the 100th element from the head would require us to iterate through the first elements and reach the 100th element which is a constant number of operations and independent of the length of the list.

As we have a pointer to the last element of the list we can delete the last element but we need to update the next pointer of the second last node/element as NULL as well, this requires us to traverse the list upto the second last element and hence its time complexity is $O(n)$.

First we need to determine the length of the list and then we need to get to the middle of the list with the help of the list length, hence the time required is $O(n)$.

The first element can be accessed in $O(1)$ time but accessing the second last element would require $O(n)$ time, hence the time required is $O(n)$.

59)Solution: (D)

**2 Subjects are compulsory so we have to keep them fixed remaining
3 subjects can be selected from 9-2 i.e. 7 subjects in $C(7,3)=35$ ways.**

60.)Solution: (10)

a=0, initially

test(24) =

i = 24, x = 0

In for loop

j= 1 , x = 0 + 24 mod 10 = 4

i = 2, a= 4

j = 2, x = 4 + 2 mod 10 = 6

i =2/10= 0 , a = 4 + 6 = 10

a= 10 is the answer.

61) Solution: B

**The number of ways in which the last three digits can be chosen is
 $10*10*10=1000$ only one of them is correct, therefore the
probability that the number is correct is $1/1000$.**

62) Solution: B

For six subnets = 3 bits required

Third subnet = 010

$192.123.220.10\ 000100 = 192.223.220.132$

63) SOLUTION C

**On applying column transforms $C2 \rightarrow C2-C1$ and $C3 \rightarrow C3-C1$ we get
the determinant equal to**

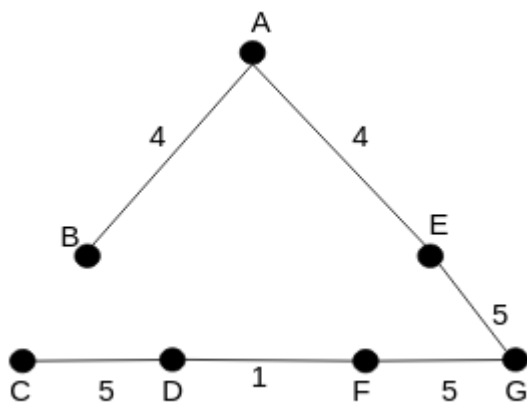
$|1\ 0\ 0\ 1\ x\ 0\ 1\ 0\ y| =$

On expanding we get xy.

64)SOLUTION A

**If we use either Prim's or Kruskal's algorithm we may get a graph
which looks like the following, please note the given graph has**

multiple MST's so it need not be the same, however the weight of the MST will remain the same.



65) Solution: (A)

S1 : TRUE.

A surjection would hit all real numbers, but this hits only the square roots of naturals. It's an injection because every natural has a different square root.

S2 :FALSE.

There are one-to-one functions that are not onto, but this statement is true if the domain and range are finite sets of the same size.

66)Solution: (A)

SMTP =25

HTTP =80

DNS =53

POP client = 110

FTP = 21

67 SOLUTION D

The binary form of $(86F)_{16}$ is : 1011 0110 1111

On converting the given binary coded decimal into Gray code we will get:

BCD: 101101101111

Gray: 111001011000

=>(E D 8)₁₆

Hence, the correct option is D

68 Solution: A

Merge sort requires additional memory so they are not inplace sorting algorithms.

69.)Solution: A(-255)

$27 \ 27 \ 2 \ * \ + \ 3 \ 4 \ * \ + \ 12 \ 7 \ * \ + \ 3 \ 12 \ 2 \ ^ \ * \ -$

Push 27, 27 and 2 onto to stack.

Pop 2 and 27 from the stack and multiply and push the result back onto the stack.

Pop 54 and 27 add them and push the result 81 back to the stack.

Push 3, 4 onto the stack

Pop 3 and 4 multiply them and push 12 to the stack.

Pop 12 and 81 add them and push 93 onto the stack.

Push 12 and 7 onto the stack.

Pop 7 and 12 multiply and push the result 84 onto the stack.

Pop 84 and 93 add them and push the result 177 onto the stack.

Push 3, 12, 2 onto the stack.

Pop 12 and 2 and evaluate $12^2=144$ and push it back onto the stack.

Pop 144 and 3 and multiply them and push the result 432 onto the stack.

Pop 432 177 and subtract them and push the result 255 onto the stack.

End of the expression is reached the result is -255

70.) Solution: B

The relationship R1 in between the entities E1, E2, E3, and E4 is many-to-many with both side partial participation. Therefore, we need a separate relation for all the 4 entities and a separate relation for the relationship R1. Thus, the total number of relations for this many-to-many relationship will be 5.

For the relationship R2 (many-to-many with one side total participation) in between entity E2 and E4, we need two separate relations for entity E2 and E4 (which we have already counted/considered above). There is no separate relation required for relationship R2 (due to the presence of composite primary key). Thus, the total number of relations required for the given ER diagram is 5.

71.) Solution: (B)

There are 12 block references in an inode, and one single-indirect block that refers to another $512/4 = 128$ blocks. So there are a total of 140 blocks. With 512 bytes per block, the maximum file size is = $140 \times 512 = 71680$ bytes.

72.) B

73.) SOLUTION A

. $\sum_d (S_d - D + 1)$ This guarantees that every diner can get all but one of the spoons it needs, with one additional spoon to guarantee that at least one diner gets all of the spoons it needs.

74.) SOLUTION C

```

// initial array A = {1,2,3};
p = A; //p points to the 0th element of array A, i.e, A[0]

p++; // p points to the 1st element of array A, i.e., A[1]

q = &p; // q points to the address where p is pointing to,
        //i.e. both p and q now point to the 1st element A[1]

p++; // p points to the 2nd element of array A, i.e., A[2],
        //since q points to the address where p is pointing to,
        //therefore, both p and q now point to the 2nd element
A[2]

(*p) = (**q)*2; //changes A[2] = A[2] * 2 --> A[2] = 3*2 = 6

// Final Array A = {1,2,6};

```

Answer : Option C) 1 2 6

75.) SOLUTION D

K-cube graph is a hypercube graph. It is bipartite, with two partitions being odd parity strings, even parity strings.

76.) SOLUTION B

Given a set of values suppose $R=\{1,2,3,4\}$ we need to find the number of relations on the set which are both partial-order and equivalence relations.

Since the relation must be reflexive, it must contain the elements $(1,1), (2,2), (3,3)$, and $(4,4)$. It is not difficult to show Hence $\{(1,1), (2,2), (3,3), (4,4)\}$ is the only relation on the set $\{1,2,3,4\}$ that is reflexive, symmetric, and anti-symmetric. Clearly it is also transitive, and hence it is the only relation that is both a partial order and an equivalence relation.

The same argument goes for any set S : The only relation that is both a partial order and an equivalence relation is the identity relation $R=\{(x,y)|x\in S\}$.

77.) Solution A

$$2^{32} \text{ blocks} \times 2^{11} \text{ bytes / block} = 2^{43} \text{ 8Terabytes.}$$

78.) SOLUTION B

$$P(\text{collision}) = P(2 \text{ or more station transmit at a time})$$

$$= 1 - (P(1 \text{ station transmit}) + P(0 \text{ station transmit}))$$

A collision occurs (i.e., with probability 1) unless either the channel is used successfully or nobody sends.

$$\text{So, } 1 - 8p(1-p)^7 - (1-p)^8.$$

79.) SOLUTION D

We will have 19 edges hence will take weights from 1 to 19.

$$1+2+3+\dots+19=19(19+1)/2=190.$$

80.) SOLUTION D

Since, G and F are disjoint, they cannot happen together.

$P(G|F)$ means Probability of event G happening given event F happened.

Since F has happened and they both cannot happen together, we know G cannot happen.

So, $P(G|F)=0$

81.) SOLUTION C

Decoder has a Multiple-input multiple-output logic circuit which converts coded inputs into coded outputs where the input and output codes are different.

82.) Solution C

Here we Apply Mean Value Theorem $[F(b)-F(a)] / (b-a) = F'(C)$ where In Interval $[a,b]$ Continuous and (a,b) Differential So Now put the value

$(7-4) / (2-(-1))$ which is Equal to $F'(c)=3/3=1$ SO C is the Answer

83.) SOLUTION D

When we evaluate post-fix expression the stack contents will be:

- 8
- 8,6
- 8,6,3
- 8,2
- 10
- 10,3
- 10,3,2
- 10,6

84.) Solution: (B)

A Hamiltonian circuit is a circuit that visits every vertex once with no repeats. Being a circuit, it must start and end at the same vertex. Graph A does not have a Hamilton circuit since there is a cut-edge, namely EG.

**Graph B has a Hamilton circuit, for example
B, A, C, E, H, G, D, F, B.**

85.) Solution: (C)

A, G, B, D, E, F, is not a valid DFS traversal as D is not a neighbouring vertex of B or cannot and all its neighbours are not yet explored, apart from that all the other options are valid DFS traversals

86.) Solution: (B)

The distance-vector routing protocol sends its complete routing table out all active interfaces at periodic time intervals. Link-state routing protocols send updates containing the state of its own links to all routers in the internetwork.

87.) SOLUTION A

Item	Profit	Weight	Profit/Weight
1	20	10	2
2	8	16	0.5
3	7	1	7
4	5	40	0.125
5	10	2	5

Since the question is referring to fractional Knapsack problem we have to be greedy in terms of profit/weight

The optimal profit can be obtained by adding item 5, 3, 1 and 7 units of item 2, the optimal profit which can be obtained is $10+7+20+3.5=40.5$

88.) Solution: (C)

205.16.37.128/28i Given block

For 4 subnets two bits require , so the sub mask = $28 + 2 = 30$

Two bits are left for host so max host per subnet= $(2^2)-2 = 2$

205.16.37.1000 00 00 = block address

Last subnet Range -> 205.16.37.1000 11 00 - 205.16.37.1000 11 11

89.) Ans c f4 is equivalent to $O(n)$

f3 is an exponential function a^n form

f2 is decreasing function which will twnds to 0

so $f2 < f4 < f1 < f3$

90.)SOLUTION D

Solution: (D)

The given POS expression of F could be represented using truth table is:

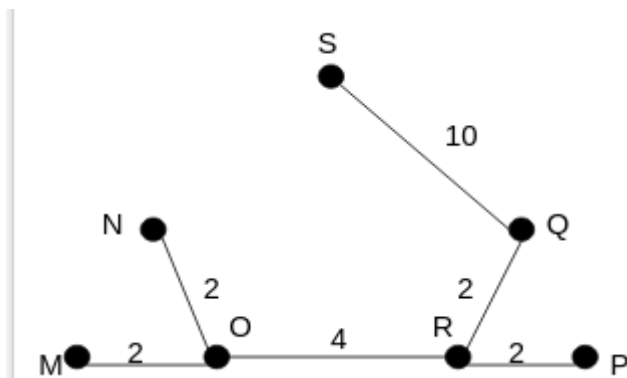
A	B	C	F	SOP representation	POS representation
0	0	0	0	$A'B'C'$	$A + B + C$
0	0	1	1	$A'B'C$	$A + B + C'$
0	1	0	1	$A'BC'$	$A + B' + C$
0	1	1	0	$A'BC$	$A + B' + C'$
1	0	0	1	$AB'C'$	$A' + B + C$
1	0	1	1	$AB'C$	$A' + B + C'$
1	1	0	1	ABC'	$A' + B' + C$
1	1	1	1	ABC	$A' + B' + C'$

The terms which are 0 in F are maxterms and the rest of the others are the minterms. Therefore, the equivalent SOP expression is:
 $A'B'C + A'BC' + AB'C' + AB'C + ABC' + ABC$.

91.) SOLUTION A

Ans a

If we apply Prim's or Kruskal's algorithm for MST we will get the MST which looks as below, the cost of the MST is $2+2+2+2+4+10=22$



The edges MP and NM are not present in every MST the above graph is an example

92.) Solution c

Inverted Page Table is the global page table which is maintained by the Operating System for all the processes. In an inverted page table, the number of entries is equal to the number of frames in the main memory.

Therefore, for Inverted Page Table, we need only the number of page frames

$$= 2^{32} / 2^{13}$$

$$= 2^{19}$$

So, number of entries in an inverted page table is 2^{19}

93.) Solution B

The maximum no of candidate keys of size r for given n attribute is nC_r . While in general, the maximum no of candidate keys for n attribute is given by ${}^nC_{\lfloor n/2 \rfloor}$. This is because the candidate keys is the minimal super keys.

Thus the maximum no of candidate keys for the given 10 attribute is ${}^6C_3 = 6! / (3! 3!) = 20$ keys

94)Solution: A

For relation E1-R1, A will be the foreign key, thus the number of foreign keys = 1.

For relation E1-E2, the relation R2 is a many-to-many relationship, thus minimum 2 foreign keys are required (A and D will be the foreign key). Thus, the number of foreign keys = 1 + 2 = 3

For relation E1-E2, the relation R3 is one-to-many, therefore, A will be the foreign key. Thus, the number of foreign keys are 4.

95.) Solution D

As we know $A \rightarrow B$ determines that A can uniquely identify B .

And $B \rightarrow A$ determine that B can uniquely identify A.

So it is one-one relationship

96.) Solution d

Since the range of 2s complement no are from -2^{n-1} to $2^{n-1}-1$ where n represent the number of bits

Therefore the maximum positive number represent by 8 bits will be $2^7-1=128-1=127$

97.) Solution B

For the CSMA/CD network

Transmission time ≥ 2 Propagation time

For the minimum frame size.

$T_t = 2 T_p$

$\text{Datasize} / \text{BW} = 2 \times (\text{Length} / \text{velocity})$

98.)SOLUTION C

The given C code snippet starts with the string "COURSE". It then moves the pointer to the second character ('O'), increments its ASCII value by 1 (changing 'O' to 'P'), and finally prints the string from that point onward.

So, the output is "PURSE".

99.) Solution B

Applying Kruskal's algorithm and then

1. We add the edges AB, BC
2. Then we add the edges FE, CD, GH.
3. Then we add the edges AF, FI, IJ
4. Then we add the edge DL.
5. Then the edge LK is added
6. Then the edge KJ is added.

Now addition of any other edge results in a cycle. Total cost is

$=2*1+3*2+3*3+1*4+ 1*5+ 1*7=33$

100.)solution c

Explanation:

Deadlock avoidance requires far too much overhead.

Every time a resource is requested, this complicated algorithm examines multiple scenarios for future use of resources to determine whether or not deadlock could possibly occur if the resource request is granted. Further, it requires that the maximum resource requirement for each process is known in advance, a maximum requirement that may or may not occur depending on which path in the program follows through the code.

A large time quantum will reduce the overhead of context switching since interrupts will be generated with relatively long intervals, hence there will be fewer interrupts. However, a short job will have to wait longer time on the ready queue before it can get to execute on the processor.

With a short time quantum, such a short job will finish quicker and produces the result to the end user